

University of Glasgow

Friday 9 May 2026

14:00 – 15:00 BST

Duration: 1 hour | CLOSED BOOK

DEGREE OF MSc

Deep Learning for MSc COMPSCI5103

Answer BOTH Questions. Total: 40 marks. CLOSED BOOK — Only the attached Crib Sheet is permitted.

CRIB SHEET (Provided — refer as needed)

Sigmoid: $a(z) = 1/(1+\exp(-z))$ $da/dz = a(1-a)$

Tanh: $a(z) = (\exp(z)-\exp(-z))/(\exp(z)+\exp(-z))$ $da/dz = (1-a^2)$

ReLU: $a(z) = \max(0, z)$ $da/dz = 1$ if $z \geq 0$, else 0

Conv/Pool output size: $W_{out} = \text{floor}((W_{in} - F + 2P) / S) + 1$

Question 1 [20 marks]

You should answer the following questions by examining the PyTorch model below.

```
class AudioNet(nn.Module):
    def __init__(self):
        super(AudioNet, self).__init__()
        self.conv1 = nn.Conv1d(in_channels=4, out_channels=8,
                                kernel_size=5, stride=1, padding=2)
        self.conv2 = nn.Conv1d(in_channels=8, out_channels=4,
                                kernel_size=3, stride=2, padding=1)
        self.fc1 = nn.Linear(4 * 25, 32)
        self.fc2 = nn.Linear(32, 1)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = F.relu(self.conv2(x))
        x = x.view(x.size(0), -1)
        x = F.relu(self.fc1(x))
        x = torch.sigmoid(self.fc2(x))
        return x

criterion = nn.BCELoss()
optimizer = optim.SGD(model.parameters(), lr=0.05,
                        momentum=0.85, weight_decay=0.005)
```

(a)

The input tensor has shape `torch.size([8, 4, 50])`.

(i) What does each of the three dimensions represent? [2]

(ii) What type of real-world data and machine learning task does this network represent? Justify your answer by referencing the architecture. [2]

[4 marks total]

(b)

Calculate the **total number of parameters** in this network. Show your working for each layer separately, stating the formula used.

*Hint: Conv1d(in, out, k) has $(k * in * out)$ weights + out biases.*

[5 marks]

(c)

Trace the tensor shape through every layer of the *forward* method for an input of shape **[8, 4, 50]**. Write the layer name and resulting tensor shape at each step.

[4 marks]

(d)

The model uses **BCELoss** with a **sigmoid** output.

(i) Is this the correct loss function for this output? Would BCEWithLogitsLoss be equivalent? Briefly explain. [2]

(ii) Describe the shape and datatype of the ground truth target tensor required by BCELoss here. [1]

[3 marks total]

(e)

The SGD optimizer uses **momentum=0.85**.

(i) Write the two update equations that define momentum-based SGD. [1]

(ii) Describe a specific type of loss surface where momentum gives a significant advantage over plain SGD. Explain why. [2]

[3 marks total]

Question 2 [20 marks]

(a)

The following **2×3 kernel** is applied to the **4×6 image** shown.

Kernel (2 rows x 3 cols):

1	0	-1
1	0	-1

Image (4 rows x 6 cols):

3	3	0	0	3	3
3	3	0	0	3	3
3	3	0	0	3	3
3	3	0	0	3	3

- (i) What type of image feature does this kernel detect? [1]
- (ii) Calculate the output with **no padding, stride=1**. Show working for all positions. [3]
- (iii) Apply **MaxPool2d(size=2x2, stride=2)** to your result from (ii). Write the output values. [2]

[6 marks total]

(b)

The figure below describes a small neural network with **sigmoid activations**. Let $\mathbf{x}$ in $\mathbb{R}^2$, $\mathbf{z}$ in $\mathbb{R}^2$, $\mathbf{h}$ in $\mathbb{R}^2$ be vectors. Let $\mathbf{M}$ in $\mathbb{R}^{2 \times 2}$ and $\mathbf{N}$ in $\mathbb{R}^{1 \times 2}$ be matrices.

The network equations are:

$$\begin{aligned} \mathbf{z} &= \mathbf{M} * \mathbf{x} \\ \mathbf{h} &= \text{sigmoid}(\mathbf{z}) \\ \mathbf{y} &= \mathbf{N} * \mathbf{h} \end{aligned}$$

- (i) Write the matrix $\mathbf{N}$ using symbolic weight notation $w_1^{(2)}$ and $w_2^{(2)}$. Also write y as an explicit sum. [2]
- (ii) Calculate the gradient vector $\text{grad}_{\mathbf{h}} \mathbf{y}$. Explain why this is simply $\mathbf{N}^T$. [2]
- (iii) Using backpropagation, determine $\text{grad}_{\mathbf{z}} \mathbf{y}$. Show the Jacobian matrix of $\mathbf{h}$ with respect to $\mathbf{z}$ and the matrix multiplication. [3]
- (iv) Explain what happens to $\text{grad}_{\mathbf{z}} \mathbf{y}$ when sigmoid is deeply saturated (h near 0 or h near 1). What problem does this cause in deep networks? [2]

[9 marks total]

(c)

The GAN training objective from Goodfellow et al. (2014) is:

$$\min_G \max_D V(D, G) = \mathbb{E}[\log D(\mathbf{x})] + \mathbb{E}[\log(1 - D(G(\mathbf{z})))]$$

- (i) Briefly describe the roles of G and D in this equation. What does each network try to do? [2]
- (ii) Explain the meaning of $D(G(\mathbf{z}))$. What value does D want this to be, and what value does G want it to be? [1]
- (iii) State one practical training instability problem that GANs suffer from and briefly explain why it occurs. [2]

[5 marks total]

END OF QUESTION PAPER